Chapter 7: G-codes

7.1 Introduction

This chapter gives detailed descriptions of the G-codes that you use to program your machine.

7.1.1 List of G-codes

CAUTION: *The sample programs in this manual have been tested for accuracy, but they are for illustrative purposes only. The programs do not define tools, offsets, or materials. They do not describe workholding or other fixturing. If you choose to run a sample program on your machine, do so in Graphics mode. Always follow safe machining practices when you run an unfamiliar program.*

NOTE: *The sample programs in this manual represent a very conservative programming style. The samples are intended to demonstrate safe and reliable programs, and they are not necessarily the fastest or most efficient way to operate a machine. The sample programs use G-codes that you might choose not to use in more efficient programs.*

Code	Description	Group	Page
G00	Rapid Motion Positioning	01	**263**
G01	Linear Interpolation Motion	01	**264**
G02	CW Circular Interpolation Motion	01	**271**
G03	CCW Circular Interpolation Motion	01	**271**
G04	Dwell	00	**273**
G09	Exact Stop	00	**274**
G10	Set Offsets	00	**274**
G14	Secondary Spindle Swap	17	**275**

Code	Description	Group	Page
G15	Secondary Spindle Swap Cancel	17	**275**
G17	XY Plane	02	**275**
G18	XZ Plane	02	**275**
G19	YZ Plane	02	**275**
G20	Select Inches	06	**276**
G21	Select Metric	06	**276**
G28	Return To Machine Zero Point	00	**276**
G29	Return From Reference Point	00	**276**
G31	Skip Function	00	**276**
G32	Thread Cutting	01	**277**
G40	Tool Nose Compensation Cancel	07	**280**
G41	Tool Nose Compensation (TNC) Left	07	**281**
G42	Tool Nose Compensation (TNC) Right	07	**281**
G50	Spindle Speed Limit	00	**281**
G50	Set Global coordinate Offset FANUC	00	**282**
G52	Set Local Coordinate System FANUC	00	**282**
G53	Machine Coordinate Selection	00	**282**
G54	Coordinate System #1 FANUC	12	**282**
G55	Coordinate System #2 FANUC	12	**282**
G56	Coordinate System #3 FANUC	12	**282**
G57	Coordinate System #4 FANUC	12	**282**
G58	Coordinate System #5 FANUC	12	**282**
G59	Coordinate System #6 FANUC	12	**282**

Code	Description	Group	Page
G61	Exact Stop Modal	15	**283**
G64	Exact Stop Cancel G61	15	**283**
G65	Macro Subprogram Call Option	00	**283**
G70	Finishing Cycle	00	**283**
G71	O.D./I.D. Stock Removal Cycle	00	**284**
G72	End Face Stock Removal Cycle	00	**287**
G73	Irregular Path Stock Removal Cycle	00	**291**
G74	End Face Grooving Cycle	00	**293**
G75	O.D./I.D. Grooving Cycle	00	**296**
G76	Threading Cycle, Multiple Pass	00	**299**
G80	Canned Cycle Cancel	09	**302**
G81	Drill Canned Cycle	09	**303**
G82	Spot Drill Canned Cycle	09	**303**
G83	Normal Peck Drilling Canned Cycle	09	**305**
G84	Tapping Canned Cycle	09	**307**
G85	Boring Canned Cycle	09	**310**
G86	Bore and Stop Canned Cycle	09	**311**
G89	Bore and Dwell Canned Cycle	09	**311**
G90	O.D./I.D. Turning Cycle	01	**312**
G92	Threading Cycle	01	**313**
G94	End Facing Cycle	01	**315**
G95	Live Tooling Rigid Tap (Face)	09	**316**
G96	Constant Surface Speed On	13	**317**

Code	Description	Group	Page
G97	Constant Surface Speed Off	13	**317**
G98	Feed Per Minute	10	**317**
G99	Feed Per Revolution	10	**317**
G100	Disable Mirror Image	00	**318**
G101	Enable Mirror Image	00	**318**
G103	Limit Block Lookahead	00	**318**
G105	Servo Bar Command	09	**319**
G110	Coordinate System #7	12	**319**
G111	Coordinate System #8	12	**319**
G112	XY to XC Interpolation	04	**317**
G113	Cancel G112	04	**320**
G114	Coordinate System #9	12	**321**
G115	Coordinate System #10	12	**321**
G116	Coordinate System #11	12	**321**
G117	Coordinate System #12	12	**321**
G118	Coordinate System #13	12	**321**
G119	Coordinate System #14	12	**321**
G120	Coordinate System #15	12	**321**
G121	Coordinate System #16	12	**321**
G122	Coordinate System #17	12	**321**
G123	Coordinate System #18	12	**321**
G124	Coordinate System #19	12	**321**
G125	Coordinate System #20	12	**321**

Code	Description	Group	Page
G126	Coordinate System #21	12	**321**
G127	Coordinate System #22	12	**321**
G128	Coordinate System #23	12	**321**
G129	Coordinate System #24	12	**321**
G154	Select Work Coordinates P1-99	12	**321**
G184	Reverse Tapping Canned Cycle For Left Hand Threads	09	**323**
G186	Reverse Live Tool Rigid Tap (For Left Hand Threads)	09	**323**
G187	Accuracy Control	00	**324**
G195	Forward Live Tool Radial Tapping (Diameter)	09	**325**
G196	Reverse Live Tool Radial Tapping (Diameter)	09	**325**
G198	Disengage Synchronous Spindle Control	00	**315**
G199	Engage Synchronous Spindle Control	00	**326**
G200	Index on the Fly	00	**328**
G211	Manual Tool Setting	-	**330**
G212	Auto Tool Setting	-	**330**
G241	Radial Drill Canned Cycle	09	**331**
G242	Radial Spot Drill Canned Cycle	09	**332**
G243	Radial Normal Peck Drilling Canned Cycle	09	**334**
G245	Radial Boring Canned Cycle	09	**336**
G246	Radial Bore and Stop Canned Cycle	09	**337**
G249	Radial Bore and Dwell Canned Cycle	09	**340**

Introduction to G-codes

G-codes are used to command specific actions for the machine: such as simple machine moves or drilling functions. They also command more complex features which can involve optional live tooling and the C Axis.

Each G-code has a group number. Each group of codes contains commands for a specific subject. For example, Group 1 G-codes command point-to point moves of the machine axes, Group 7 are specific to the Cutter Compensation feature.

Each group has a dominant G-code; referred to as the default G-code. A default G-code means they are the one in each group the machine uses unless another G-code from the group is specified. For example programming an X, Z move like this, `X-2. Z-4.` will position the machine using `G00`.

NOTE: *Proper programming technique is to preface all moves with a G-code.*

Default G-codes for each group are shown on the **`Current Commands`** screen under**`All Active Codes`**. If another G-code from the group is commanded (active), that G-code is displayed on the **`All Active Codes`** screen.

G-code commands are either modal or non-modal. A modal G-code stays in effect until the end of the program or until you command another G-code from the same group. A non-modal G-code affects only the line it is in; it does not affect the next program line. Group 00 codes are non-modal; the other groups are modal.

NOTE: *The Haas Intuitive Programming System (IPS) is a programming mode that either hides G-codes or completely bypasses the use of G-codes.*

Canned Cycles

Canned cycles simplify part programming. Most common Z-axis repetitive operations, such as drilling, tapping, and boring, have canned cycles. When active, a canned cycle executes at every new axis position. Canned cycles execute axis motions as rapid commands (`G00`) and the canned cycle operation is performed after the axis motion. This applies to `G17`, `G19` cycles, and Y-Axis movements on Y-Axis lathes.

Using Canned Cycles

Modal canned cycles stay in effect after you define them, and they execute in the Z-axis for each position of the X, Y, and C-Axes.

NOTE: *X, Y, or C-Axis positioning moves during a canned cycle are rapid moves.*

Canned cycles operate differently, depending on whether you use incremental (U,W) or absolute (X, Y, or C) positions.

If you define a loop count (`Lnn` code number) in the canned cycle block, the canned cycle repeats that many times with an incremental (`U` or `W`) move between each cycle.

Enter the number of repeats (`L`) each time you want to repeat a canned cycle. The control does not remember the number of repeats (`L`) for the next canned cycle.

You should not use spindle control M-codes while a canned cycle is active.

Canceling a Canned Cycle

`G80` cancels all canned cycles. `G00` or `G01` code also cancel a canned cycle. A canned cycle stays active until `G80`, `G00`, or `G01` cancels it.

Canned Cycles with Live Tooling

The canned cycles `G81`, `G82`, `G83`, `G85`, `G86`, `G87`, `G88`, `G89`, `G95`, and `G186` can be used with axial live tooling, and `G241`, `G242`, `G243`, `G245`, and `G249` can be used with radial live tooling. Some programs must be checked to be sure they turn on the main spindle before running the canned cycles.

NOTE: *`G84` and G184 are not usable with live tooling.*

G00 Rapid Motion Positioning (Group 01)

*__B__ - B-axis motion command
*__C__ - C-Axis motion command
*__U__ - X-axis incremental motion command
*__W__ - Z-axis incremental motion command
*__X__ - X-axis absolute motion command
*__Y__ - Y-axis absolute motion command
*__Z__ - Z-axis absolute motion command

* indicates optional

This G code is used to move the machine's axes at the maximum speed. It is primarily used to quickly position the machine to a given point before each feed (cutting) command. This G code is modal, so a block with `G00` causes all following blocks to be rapid motions until another cutting move is specified.

NOTE: *Generally, rapid motion will not be in a straight line. Each axis specified is moved at the same speed, but all axes will not necessarily complete their motions at the same time. The machine will wait until all motions are complete before starting the next command.*

G01 Linear Interpolation Motion (Group 01)

F - Feed rate
* **B** - B-axis motion command
* **C** - C-Axis motion command
* **U** - X-axis incremental motion command
* **W** - Z-axis incremental motion command
* **X** - X-axis absolute motion command
* **Y** - Y-axis absolute motion command
* **Z** - Z-axis absolute motion command
* **A** - Optional angle of movement (used with only one of `X`, `Z`, `U`, `W`)
* **I** - X-axis chamfering from Z to X (the sign does not matter, only for 90 degree turns)
* **K** - Z-axis chamfering from X to Z (the sign does not matter, only for 90 degree turns)

* **,C** - Distance from center of intersection where the chamfer begins (the sign does not matter, can chamfer non-90 degree lines)
* **,R / R** - Radius of the fillet or arc (the sign does not matter)

This G code provides for straight line (linear) motion from point to point. Motion can occur in 1 or more axes. You can command a `G01` with 3 or more axes All axes will start and finish motion at the same time. The speed of all axes is controlled so that the feed rate specified is achieved along the actual path. The C-Axis may also be commanded and this will provide a helical (spiral) motion. A C-Axis feed rate is dependent on the C-Axis diameter setting (Setting 102) to create a helical motion. The `F` address (feedrate) command is modal and may be specified in a previous block. Only the axes specified are moved.

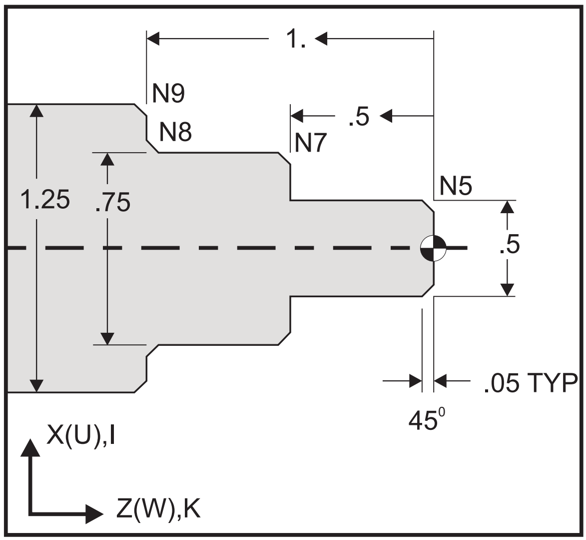
Corner Rounding and Chamfering Example

A chamfer block or a corner rounding block can be automatically inserted between two linear interpolation blocks by specifying `,C` (chamfering) or `,R` (corner rounding).

NOTE: *Both of these variables use a comma symbol (,) before the variable.*

There must be a terminating linear interpolation block after the beginning block (a `G04` pause may intervene). These two linear interpolation blocks specify a theoretical corner of intersection. If the beginning block specifies a `,C` (comma C) the value after the `C` is the distance from the corner of intersection where the chamfer begins and also the distance from that same corner where the chamfer ends. If the beginning block specifies a `,R` (comma R) the value after the R is the radius of a circle tangent to the corner at two points: the beginning of the corner rounding arc block that is inserted and the endpoint of that arc. There can be consecutive blocks with chamfer or corner rounding specified. There must be movement on the two axes specified by the selected plane (the active plane X-Y (`G17`), X-Z (`G18`) or Y-Z (`G19`). For chamfering a 90° angle only, an `I` or `K` value can be substituted where `,C` is used.

F7.1: Chamfering




```
%
o60011 (G01 CHAMFERING) ;
(G54 X0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is an OD cutting tool) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G50 S1000 (Limit spindle to 1000 RPM) ;
G97 S500 M03 (CSS off, Spindle on CW) ;
G00 G54 X0 Z0.25 (Rapid to 1st position) ;
M08 (Coolant on) ;
(BEGIN CUTTING BLOCKS) ;
G01 Z0 F0.005 (Feed to Z0) ;
N5 G01 X0.50 K-0.050 (Chamfer 1) ;
G01 Z-0.5 (Linear feed to Z-0.5) ;
```

```
N7 G01 X0.75 K-0.050 (Chamfer 2) ;
N8 G01 Z-1.0 I0.050 (Chamfer 3) ;
N9 G01 X1.25 K-0.050 (Chamfer 4) ;
G01 Z-1.5 (Feed to Z-1.5) ;
(BEGIN COMPLETION BLOCKS) ;
G00 X1.5 M09 (Rapid Retract, Coolant off) ;
G53 X0 (X home) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;
%
```

This G-code syntax automatically includes a 45° chamfer or corner radius between two blocks of linear interpolation which intersect a right (90 degree) angle.

Chamfering Syntax

```
G01 X(U) x Kk ;
G01 Z(W) z Ii ;
```

Corner Rounding Syntax

```
G01 X(U) x Rr ;
G01 Z(W) z Rr ;
```

Addresses:

I = chamfering, Z to X

K = chamfering, X to Z

R = corner rounding (X or Z axis direction)

Notes:

1. Incremental programming is possible if `U` or `W` is specified in place of `X` or `Z`, respectively. So its actions are as follows:
   X(current position + i) = Ui

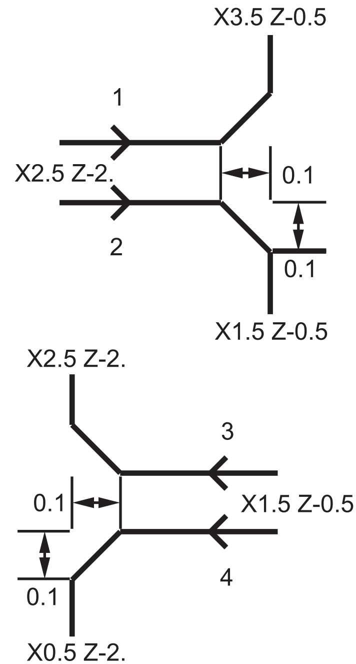
   Z(current position + k) = Wk

   X(current position + r) = Ur

   Z(current position + r) = Wr
2. Current position of X or Z Axis is added to the increment.
3. `I`, `K` and `R` always specify a radius value (radius programming value).

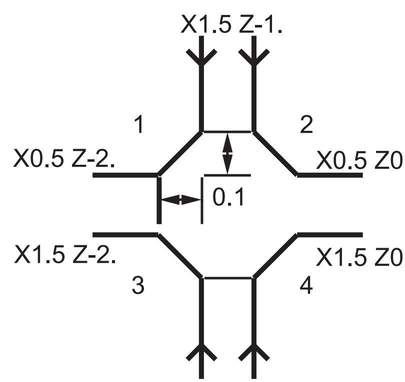
F7.2: Chamfering Code Z to X: [A] Chamfering, [B] Code/Example, [C] Movement.

A	B	C
1. Z+ to X+	X2.5 Z-2; G01 Z-0.5 I0.1; X3.5;	X2.5 Z-2; G01 Z-0.6; X2.7 Z-0.5; X3.5;
2. Z+ to X-	X2.5 Z-2.; G01 Z-0.5 I-0.1; X1.5;	X2.5 Z-2.; G01 Z-0.6; X2.3 Z-0.5; X1.5;
3. Z- to X+	X1.5 Z-0.5.; G01 Z-2. I0.1; X2.5;	X1.5 Z-0.5 G01 Z-1.9; X1.7 Z-2.; X2.5;
4. Z- to X-	X1.5 Z-0.5.; G01 Z-2. I-0.1; X0.5;	X1.5 Z-0.5; G01 Z-1.9; X1.3 Z-2. X0.5;



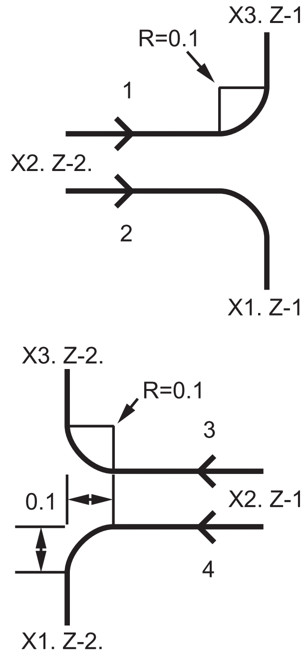
F7.3: Chamfering Code X to Z: [A] Chamfering, [B] Code/Example, [C] Movement.

A	B	C
1. X- to Z-	X1.5 Z-1.; G01 X0.5 K-0.1; Z-2.;	X1.5 Z-1.; G01 X0.7; X0.5 Z-1.1; Z-2.
2. X- to Z+	X1.5 Z-1.; G01 X0.5 K0.1; Z0.;	X1.5 Z-1.; G01 X0.7; X0.5 Z-0.9; Z0.;
3. X+ to Z-	X0.5 Z-1.; G01 X1.5 K-0.1; Z-2.;	X0.5 Z-1.; G01 X1.3; X1.5 Z-1.1; Z-2.
4. X+ to Z+	X0.5 Z-1.; G01 X1.5 K0.1; Z0.;	X0.5 Z-1.; G01 X1.3; X1.5 Z-0.9; Z0.;



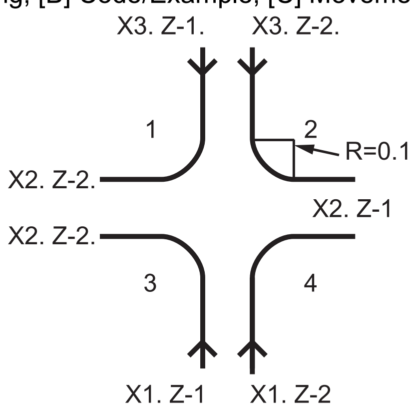
F7.4: Corner Rounding Code Z to X: [A] Corner rounding, [B] Code/Example, [C] Movement.

A	B	C
1. Z+ to X+	X2. Z-2.; G01 Z-1 R.1; X3.;	X2. Z-2.; G01 Z-1.1; G03 X2.2 Z-1. R0.1; G01 X3.;
2. Z+ to X-	X2. Z-2.; G01 Z-1. R-0.1; X1.;	X2. Z-2.; G01 Z-1.1; G02 X1.8 Z-1 R0.1; G01 X1.;
3. Z- to X+	X2. Z-1.; G01 Z-2. R0.1; X3.;	X2. Z-1.; G01 Z-1.9; G02 X2.2 Z-2. R0.1; G01 X3.;
4. Z- to X-	X2. Z-1.; G01 Z-2. R-0.1; X1.;	X2. Z-1.; G01 Z-1.9. ; G03 X1.8 Z-2.; G01 X1.;



F7.5: Corner Rounding Code X to Z: [A] Corner rounding, [B] Code/Example, [C] Movement.

A	B	C
1. X- to Z-	X3. Z-1.; G01 X0.5 R-0.1; Z-2.;	X3. Z-1.; G01 X0.7; X0.5 Z-1.1; Z-2.
2. X- to Z+	X3. Z-2.; G01 X0.5 R0.1; Z0.;	X3. Z-2.; G01 X0.7; X0.5 Z-0.9; Z0.;
3. X+ to Z-	X1. Z-1.; G01 X1.5 R-0.1; Z-2.;	X1. Z-1.; G01 X1.3; X1.5 Z-1.1; Z-2.
4. X+ to Z+	X1. Z-2.; G01 X1.5 R0.1; Z0.;	X1. Z-21.; G01 X1.3; X1.5 Z-0.9; Z0.;



Rules:

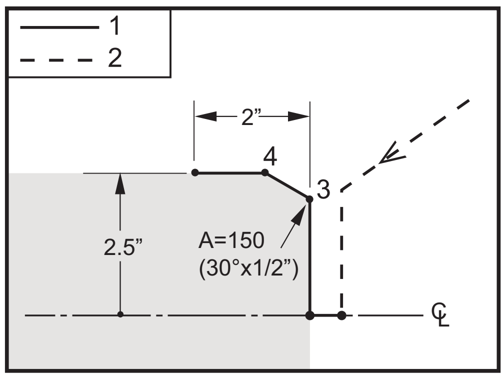
1. Use `K` address only with `X(U)` address. Use `I` address only with `Z(W)` address.
2. Use `R` address with either `X(U)` or `Z(W)`, but not both in the same block.

3. Do not use I and K together on the same block. When using R address, do not use I or K.
4. The next block must be another single linear move that is perpendicular to the previous one.
5. Automatic chamfering or corner rounding cannot be used in a threading cycle or in a Canned cycle.
6. Chamfer or corner radius must be small enough to fit between the intersecting lines.
7. Use only a single X or Z-axis move in linear mode (G01) for chamfering or corner rounding.

G01 Chamfering with A

When specifying an angle (A), command motion in only one of the other axes (X or Z), the other axis is calculated based on the angle.

F7.6: G01 Chamfering with A: [1] Feed, [2] Rapid, [3] Start Point, [4] Finish Point.



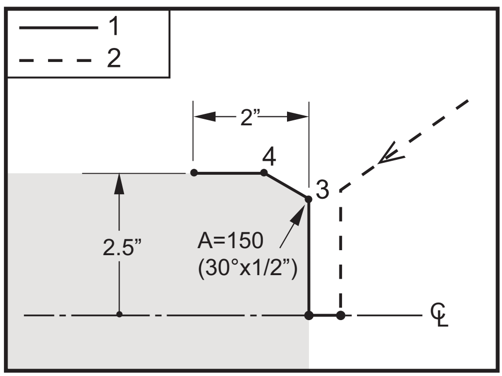

```
%
o60012 (G01 CHAMFERING WITH 'A') ;
(G54 X0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is an OD cutting tool) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G50 S1000 (Limit spindle to 1000 RPM) ;
G97 S500 M03 (CSS off, Spindle on CW) ;
G00 G54 X4. Z0.1 (Rapid to clear position) ;
M08 (Coolant on) ;
X0 (Rapid to center of diameter) ;
(BEGIN CUTTING BLOCKS) ;
G01 Z0 F0.01 (Feed towards face) ;
G01 X4. (position 3) ;
X5. A150. (position 4) ;
```

```
Z-2. (Feed to back of part) ;
(BEGIN COMPLETION BLOCKS) ;
G00 X6. M09 (Rapid Retract, Coolant off) ;
G53 X0 (X home) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;
%
```

NOTE: *A -30 = A150; A -45 = A135*

When specifying an angle (`A`), command motion in only one of the other axes (X or Z), the other axis is calculated based on the angle.

F7.7: `G01` Chamfering with A: [1] Feed, [2] Rapid, [3] Start Point, [4] Finish Point.




```
%
o60012 (G01 CHAMFERING WITH 'A') ;
(G54 X0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is an OD cutting tool) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G50 S1000 (Limit spindle to 1000 RPM) ;
G97 S500 M03 (CSS off, Spindle on CW) ;
G00 G54 X4. Z0.1 (Rapid to clear position) ;
M08 (Coolant on) ;
X0 (Rapid to center of diameter) ;
(BEGIN CUTTING BLOCKS) ;
G01 Z0 F0.01 (Feed towards face) ;
G01 X4. (position 3) ;
X5. A150. (position 4) ;
```

```
Z-2. (Feed to back of part) ;
(BEGIN COMPLETION BLOCKS) ;
G00 X6. M09 (Rapid Retract, Coolant off) ;
G53 X0 (X home) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;
%
```

NOTE: *A -30 = A150; A -45 = A135*

G02 CW/G03 CCW Circular Interpolation Motion (Group 01)

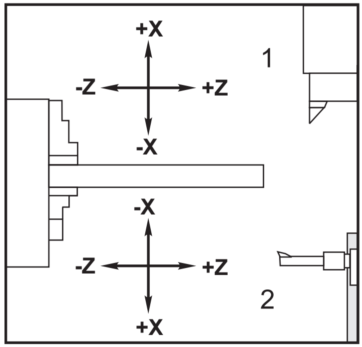
F - Feed rate
*__I__ - Distance along X-axis to center of circle
*__J__ - Distance along Y-axis to center of circle
*__K__ - Distance along Z-axis to center of circle
*__R__ - Radius of arc
*__U__ - X-axis incremental motion command
*__W__ - Z-axis incremental motion command
*__X__ - X-axis absolute motion command
*__Y__ - Y-axis absolute motion command
*__Z__ - Z-axis absolute motion command

* indicates optional

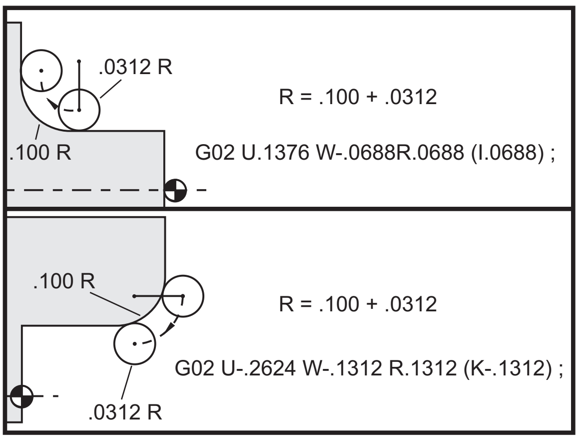
These G codes are used to specify a circular motion (CW or CCW) of the linear axes (Circular motion is possible in the X and Z axes as selected by `G18`). The `X` and `Z` values are used to specify the end point of the motion and can use either absolute (`X` and `Z`) or incremental motion (`U` and `W`). If either the `X` or `Z` is not specified, the endpoint of the arc is the same as the starting point for that axis. There are two ways to specify the center of the circular motion; the first uses `I` or `K` to specify the distance from the starting point to the center of the arc; the second uses `R` to specify the radius of the arc.

For information on `G17` and `G19` Plane Milling, see the Live Tooling section.

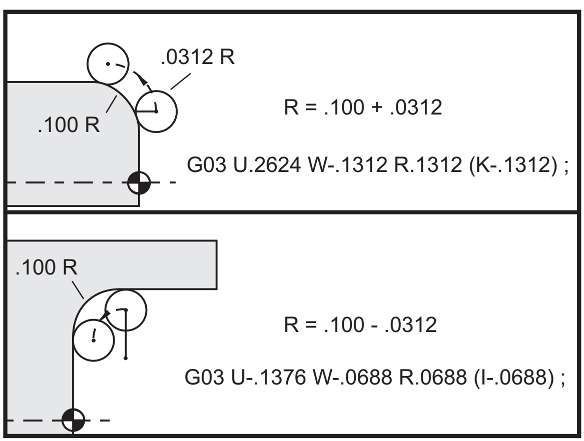
F7.8: `G02` Axis Definitions: [1] Turret Lathes, [2] Table Lathes.



F7.9: `G02` and `G03` Programs



G02



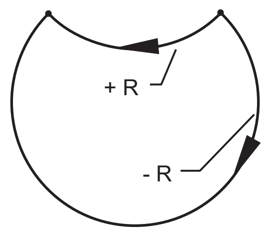
G03

`R` is used to specify the radius of the arc. With a positive `R`, the control will generate a path of 180 degrees or less; to generate a radius of over 180 degrees, specify a negative `R`. `X` or `Z` is required to specify an endpoint if different from the starting point.

The following lines cut an arc of less than 180 degrees:

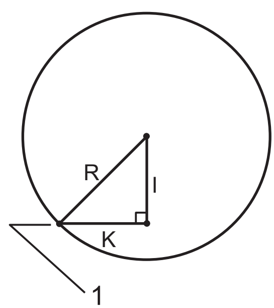
```
G01 X3.0 Z4.0 ;
G02 Z-3.0 R5.0 ;
```

F7.10: G02 Arc Using Radius



`I` and `K` are used to specify the center of the arc. When `I` and `K` are used, `R` may not be used. The `I` or `K` is the signed distance from the starting point to the center of the circle. If only one of `I` or `K` is specified, the other is assumed to be zero.

F7.11: G02 Defined X and Z: [1] Start.



G04 Dwell (Group 00)

P - The dwell time in seconds or milliseconds

NOTE: *The P values are modal. This means if you are in the middle of a canned cycle and a `G04 Pnn` or an `M97 Pnn` is used the P value will be used for the dwell / subprogram as well as the canned cycle.*

G04 specifies a delay or dwell in the program. The block with G04 delay for the time specified by the P address code. For example:

```
G04 P10.0. ;
```

Delays the program for 10 seconds.

NOTE: *G04 P10. is a dwell of 10 seconds; G04 P10 is a dwell of 10 milliseconds. Make sure you use decimal points correctly so that you specify the correct dwell time.*

G09 Exact Stop (Group 00)

G09 code is used to specify a controlled axes stop. It affects only the block in which it is commanded. It is non-modal and does not affect the blocks that come after the block where it is commanded. Machine moves decelerate to the programmed point before the control processes the next command.

G10 Set Offsets (Group 00)

G10 lets you set offsets within the program. G10 replaces manual offset entry (i.e. Tool length and diameter, and work coordinate offsets).

L - Selects offset category.

- L2 Work coordinate origin for COMMON and G54-G59
- L10 Geometry or shift offset
- L1 or L11 Tool wear
- L20 Auxiliary work coordinate origin for G110-G129

P - Selects a specific offset.

- P1-P50 - References geometry, wear or work offsets (L10-L11)
- P0 - References COMMON work coordinate offset (L2)
- P1-P6 - G54-G59 references work coordinates (L2)
- P1-P20 G110-G129 references auxiliary coordinates (L20)
- P1-P99 G154 P1-P99 reference auxiliary coordinate (L20)

Q - Imaginary tool nose tip direction
R - Tool nose radius
***U** - Incremental amount to be added to X-axis offset
***W** - Incremental amount to be added to Z-axis offset
***X** - X-axis offset
***Z** - Z-axis offset

* indicates optional

G14 Secondary Spindle Swap / G15 Cancel (Group 17)

`G14` causes the secondary spindle to become the primary spindle, so that the secondary spindle reacts to commands normally used for the main spindle. For example, `M03`, `M04`, `M05` and `M19` affect the secondary spindle, and `M143`, `M144`, `M145`, and `M119` (secondary spindle commands) cause an alarm.

NOTE: *`G50` limits the secondary spindle speed, and `G96` sets the secondary spindle surface feed value. These G-codes adjust the secondary spindle speed when there is motion in the X Axis. `G01` Feed Per Rev feeds based on the secondary spindle.*

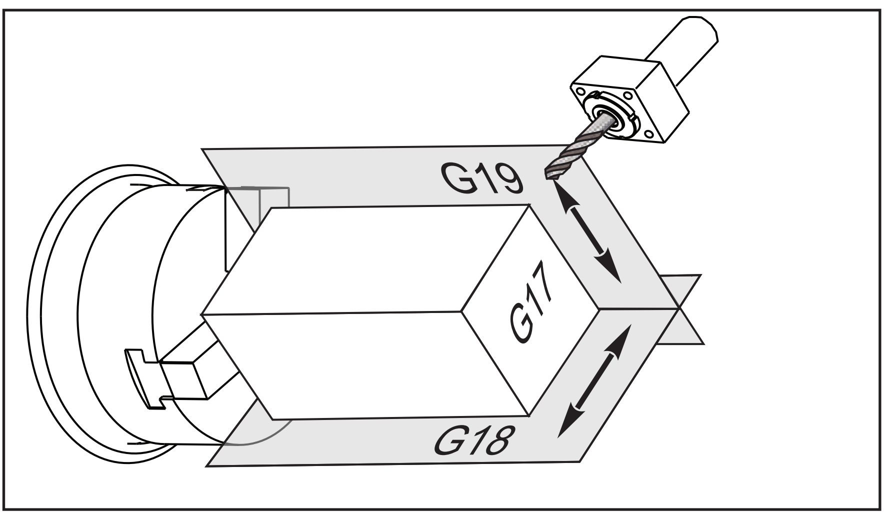
`G14` automatically activates Z-Axis mirroring. If the Z Axis is already mirrored (Setting 47 or `G101`), the mirror function is canceled.

`G14` is canceled by `G15`, `M30`, at the end of a program, or when you press **[RESET]**.

G17 XY Plane / G18 XZ Plane / G19 YZ Plane (Group 02)

This code defines the plane in which tool path motion is performed. Programming tool nose radius compensation `G41` or `G42` applies Tool Radius cutter compensation in the `G17` plane, regardless of whether `G112` is active or not. For more information, refer to Cutter Compensation in the Programming section. Plane selection codes are modal and remain in effect until another plane is selected.

F7.12: `G17`, `G18`, and `G19` Plane Selection



Program format with tool nose compensation:

```
G17 G01 X_ Y_ F_ ;
G40 G01 X_ Y_ I_ J_ F_ ;
```

G20 Select Inches / G21 Select Metric (Group 06)

Use G20 (inch) and G21 (mm) codes are to make sure that the inch/metric selection is set correctly for the program. Use Setting 9 to select between inch and metric programming. G20 in a program causes an alarm if Setting 9 is not set to inch.

G28 Return to Machine Zero Point (Group 00)

The G28 code returns all axes (X, Y, Z, B and C) simultaneously to the machine zero position when no axis is specified on the G28 line.

Alternatively, when one or more axes locations are specified on the G28 line, G28 will move to the specified locations and then to machine zero. This is called the G29 reference point; it is saved automatically for optional use in G29.

```
G28 X0 Z0 (moves to X0 Z0 in the current work coordinate system
then to machine zero) ;
G28 X1. Z1. (moves to X1. Z1. in the current work coordinate
system then to machine zero) ;
G28 U0 W0 (moves directly to machine zero because the initial
incremental move is zero) ;
G28 U-1. W-1 (moves incrementally -1. in each axis then to
machine zero) ;
```

G29 Return From Reference Point (Group 00)

G29 moves the axes to a specific position. The axes selected in this block are moved to the G29 reference point saved in G28, and then moved to the location specified in the G29 command.

G31 Feed Until Skip (Group 00)

(This G-code is optional and requires a probe.)

This G-code is used to record a probed location to a macro variable.

NOTE: *Turn on the probe before using `G31`.*

F - Feedrate in inches (mm) per minute
*__U__ - X-axis incremental motion command
*__V__ - Y-axis incremental motion command
*__W__ - Z-axis incremental motion command
X - X-axis absolute motion command
Y - Y-axis absolute motion command
Z - Z-axis absolute motion command
C - C-Axis absolute motion command

* indicates optional

This G-code moves the programmed axes while looking for a signal from the probe (skip signal). The specified move is started and continues until the position is reached or the probe receives a skip signal. If the probe receives a skip signal during the `G31` move, the control beeps and the skip signal position is recorded to macro variables. The program then executes the next line of code. If the probe does not receive a skip signal during the `G31` move, the control does not beep, the skip signal position is recorded at the end of the programmed move, and the program continues.

Macro variables `#5061` through `#5066` are designated to store skip signal positions for each axis. For more information about these skip signal variables see Macros in the Programming section of this manual.

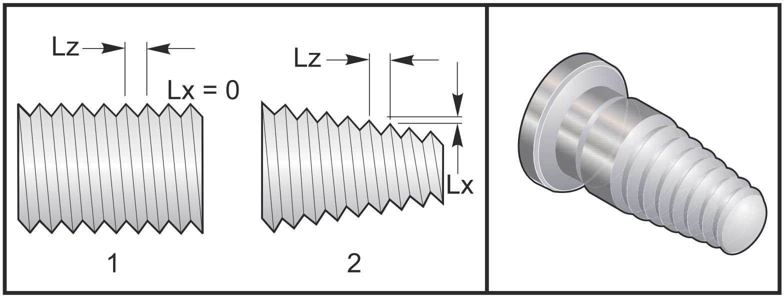
Do not use Cutter Compensation (`G41` or `G42`) with a `G31`.

G32 Thread Cutting (Group 01)

F - Feedrate in inches (mm) per minute
Q - Thread Start Angle (optional). See example on the following page.
U/W - X/Z-axis incremental positioning command. (Incremental thread depth values are user specified)
X/Z - X/Z-axis absolute positioning command. (Thread depth values are user specified)

NOTE: *Feedrate is equivalent to thread lead. Movement on at least one axis must be specified. Tapered threads have lead in both X and Z. In this case set the feedrate to the larger of the two leads. `G99` (Feed per Revolution) must be active.*

F7.13: `G32` Definition of Lead (Feedrate): [1] Straight thread, [2] Tapered thread.



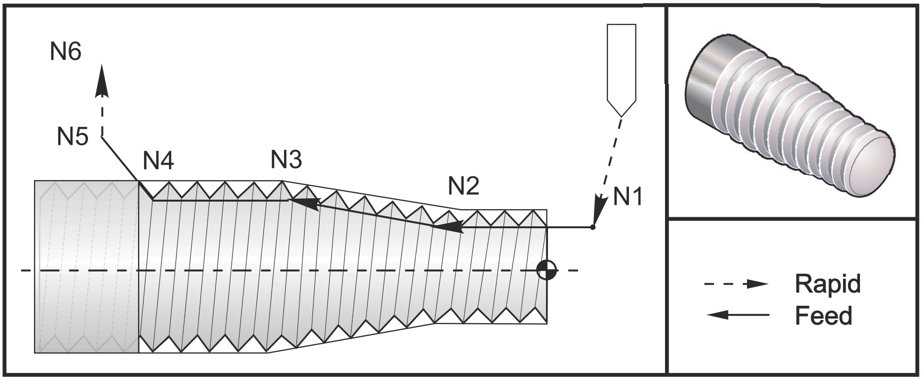
`G32` differs from other thread cutting cycles in that taper and/or lead can vary continuously throughout the entire thread. In addition, no automatic position return is performed at the end of the threading operation.

At the first line of a `G32` block of code, axis feed is synchronized with the rotation signal of the spindle encoder. This synchronization remains in effect for each line in a `G32` sequence. It is possible to cancel `G32` and recall it without losing the original synchronization. This means multiple passes will exactly follow the previous tool path. (The actual spindle RPM must be exactly the same between passes).

NOTE: *Single Block Stop and Feed Hold are deferred until last line of a `G32` sequence. Feedrate Override is ignored while `G32` is active, Actual Feedrate will always be 100% of programmed feedrate. `M23` and `M24` have no affect on a `G32` operation, the user must program chamfering if needed. `G32` must not be used with any G-code Canned Cycles (i.e.: `G71`). Do Not change spindle RPM during threading.*

CAUTION: *`G32` is Modal. Always cancel `G32` with another Group 01 G-code at the end of a threading operation. (Group 01 G-codes: `G00`, `G01`, `G02`, `G03`, `G32`, `G90`, `G92`, and `G94`.*

F7.14: Straight-to-Taper-to-Straight Thread Cutting Cycle



NOTE: *Example is for reference only. Multiple passes are usually required to cut actual threads.*

```
%
o60321 (G32 THREAD CUTTING WITH TAPER) ;
(G54 X0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is an OD thread tool) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G50 S1000 (Limit spindle to 1000 RPM) ;
G97 S500 M03 (CSS off, Spindle on CW) ;
N1 G00 G54 X0.25 Z0.1 (Rapid to 1st position) ;
M08 (coolant on) ;
(BEGIN CUTTING BLOCKS) ;
N2 G32 Z-0.26 F0.065 (Straight thread, Lead = .065) ;
N3 X0.455 Z-0.585 (Blend to tapered thread) ;
N4 Z-0.9425 (Blend back to straight thread) ;
N5 X0.655 Z-1.0425 (Pull off at 45 degrees) ;
(BEGIN COMPLETION BLOCKS) ;
N6 G00 X1.2 M09 (Rapid Retract, Coolant off) ;
G53 X0 (X home) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;
%
```

G40 Tool Nose Compensation Cancel (Group 07)

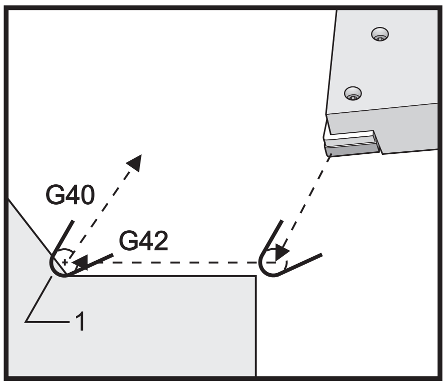
*__X__ - X Axis absolute location of departure target
*__Z__ - Z Axis absolute location of departure target
*__U__ - X Axis incremental distance to departure target
*__W__ - Z Axis incremental distance to departure target

* indicates optional

`G40` cancels `G41` or `G42`. Programming `Txx00` will also cancel tool nose compensation. Cancel tool nose compensation before the end of a program.

The tool departure usually does not correspond with a point on the part. In many cases overcutting or undercutting can occur.

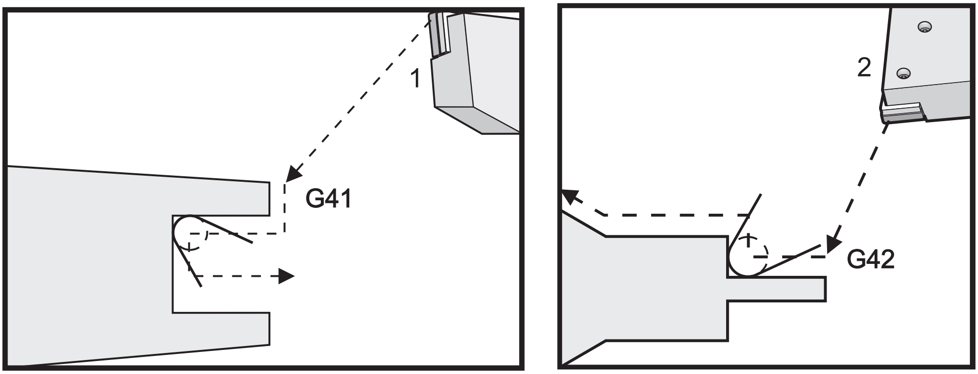
F7.15: `G40` TNC Cancel: [1] Overcut.



G41 Tool Nose Compensation (TNC) Left / G42 TNC Right (Group 07)

`G41` or `G42` will select tool nose compensation. `G41` moves the tool to the left of the programmed path to compensate for the size of a tool and vice versa for `G42`. A tool offset must be selected with a Tnnxx code. where xx corresponds to the offsets that are to be used with the tool. For more information, see Tool Nose Compensation in the Operation section of this manual.

F7.16: `G41` TNC Right and `G42` TNC Left: [1] Tip = 2, [2] Tip = 3.



G41 G42

G50 Spindle Speed Limit

`G50` can be used to limit the maximum spindle speed. The control will not allow the spindle to exceed the `S` address value specified in the `G50` command. This is used in constant surface feed mode (`G96`).

This G code will also limit the secondary spindle on DS-Series machines.

```
N1G50 S3000 (Spindle rpm will not exceed 3000 rpm) ;
N2G97 M3 (Enter constant surface speed cancel, spindle on) ;
```

NOTE: *To cancel this command, use another `G50` and specify the maximum spindle RPM for the machine.*

G50 Set Global Coordinate Offset FANUC (Group 00)

U - Incremental amount and direction to shift global X coordinate.
X - Absolute global coordinate shift.
W - Incremental amount and direction to shift global Z coordinate.
Z - Absolute global coordinate shift.
S - Limit spindle speed to specified value

`G50` performs several functions. It sets and shifts the global coordinate and it limits the spindle speed to a maximum value. Refer to the Global Coordinate System topic in the Programming section for a discussion of these.

To set the global coordinate, command `G50` with an `X` or `Z` value. The effective coordinate becomes the value specified in address code `X` or `Z`. Current machine location, work offsets, and tool offsets are taken into account. The global coordinate is calculated and set. For example:

```
G50 X0 Z0 (Effective coordinates are now zero) ;
```

To shift the global coordinate system, specify `G50` with a `U` or `W` value. The global coordinate system is shifted by the amount and direction specified in `U` or `W`. The current effective coordinate displayed changes by this amount in the opposite direction. This method is often used to place the part zero outside of the work cell. For example:

```
G50 W-1.0 (Effective coordinates are shifted left 1.0) ;
```

G52 Set Local Coordinate System FANUC (Group 00)

This code selects the user coordinate system.

G53 Machine Coordinate Selection (Group 00)

This code temporarily cancels work coordinates offsets and uses the machine coordinate system. This code will also ignore tool offsets.

G54-G59 Coordinate System #1 - #6 FANUC (Group 12)

`G54` - `G59` codes are user-settable coordinate systems, #1 - #6, for work offsets. All subsequent references to axes' positions are interpreted in the new coordinate system. Work coordinate system offsets are entered from the **`Active Work Offset`** display page. For additional offsets, refer to `G154` on page **321**.

G61 Exact Stop Mode (Group 15)

The G61 code is used to specify exact stop. Rapid and interpolated moves will decelerate to an exact stop before another block is processed. In exact stop, moves will take a longer time and continuous cutter motion will not occur. This may cause deeper cutting where the tool stops.

G64 Cancels Exact Stop Mode (Group 15)

G64 code cancels exact stop and selects the normal cutting mode.

G65 Macro Subprogram Call Option (Group 00)

G65 is described in the Macros programming section.

G70 Finishing Cycle (Group 00)

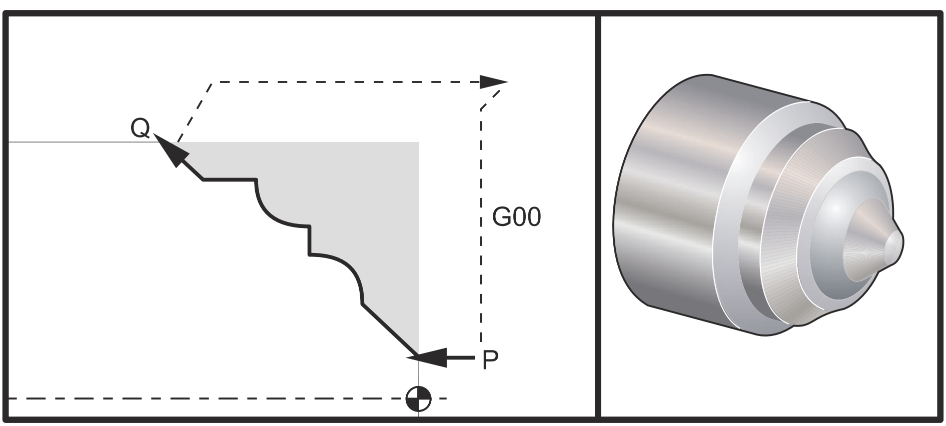
The G70 Finishing Cycle can be used to finish cut paths that are rough cut with stock removal cycles such as G71, G72 and G73.

P - Starting Block number of routine to execute
Q - Ending Block number of routine to execute

G18 Z-X plane must be active

NOTE: *The P values are modal. This means if you are in the middle of a canned cycle and a G04 Pnn or an M97 Pnn is used the P value will be used for the dwell / subprogram as well as the canned cycle.*

F7.17: G70 Finishing Cycle: [P] Starting block, [Q] Ending Block.




```
G71 P10 Q50 F.012 (rough out N10 to N50 the path) ;
```

```
N10 ;
F0.014 ;
... ;
N50 ;
... ;
G70 P10 Q50 (finish path defined by N10 to N50) ;
```

The `G70` cycle is similar to a local subprogram call. However, the `G70` requires that a beginning block number (`P` code) and an ending block number (`Q` code) be specified.

The `G70` cycle is usually used after a `G71`, `G72` or `G73` has been performed using the blocks specified by `P` and `Q`. Any `F`, `S`, or `T` codes with the `PQ` block are effective. After execution of the `Q` block, a rapid (`G00`) is executed returning the machine to the start position that was saved before the starting of the `G70`. The program then returns to the block following the `G70` call. A subprogram in the `PQ` sequence is acceptable providing that the subprogram does not contain a block with an `N` code matching the `Q` specified by the `G70` call. This feature is not compatible with FANUC controls.

After a `G70`, the block following the `G70` will be executed, not the block with an `N` code matching the `Q` code specified by the `G70` call.

G71 O.D./I.D. Stock Removal Cycle (Group 00)

First Block (Only use when using two block G71 notation)
*__U__ - Depth of cut for each pass of stock removal, positive radius
*__R__ - Retract height for each pass of stock removal
Second Block
*__D__ - Depth of cut for each pass of stock removal, positive radius (Only use when using one block G71 notation)
*__F__ - Feedrate in inches (mm) per minute (`G98`) or per revolution (`G99`) to use throughout `G71 PQ` block
*__I__ - X-axis size and direction of `G71` rough pass allowance, radius
*__K__ - Z-axis size and direction of `G71` rough pass allowance
P - Starting Block number of path to rough
Q - Ending Block number of path to rough
*__S__ - Spindle speed to use throughout `G71 PQ` block
*__T__ - Tool and offset to use throughout `G71 PQ` block
*__U__ - X-axis size and direction of `G71` finish allowance, diameter
*__W__ - Z-axis size and direction of `G71` finish allowance

* indicates optional

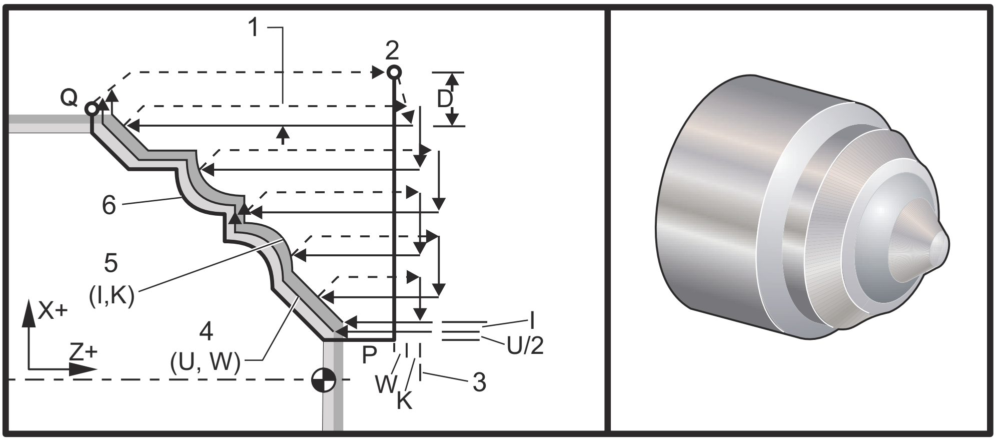
`G18 Z-X` plane must be active.

2 Block G71 Programming Example:

```
G71 U... R...
G71 F... I... K... P... Q... S... T... U... W...
```

NOTE: *The P values are modal. This means if you are in the middle of a canned cycle and a `G04 Pnn` or an `M97 Pnn` is used the P value will be used for the dwell / subprogram as well as the canned cycle.*

F7.18: `G71` Stock Removal: [1] Setting 287, [2] Start position, [3] Z-Axis clearance plane, [4] Finishing allowance, [5] Roughing allowance, [6] Programmed path.



This canned cycle roughs material on a part given the finished part shape. Define the shape of a part by programming the finished tool path and then use the `G71 PQ` block. Any `F`,`S` or `T` commands on the `G71` line or in effect at the time of the `G71` is used throughout the `G71` roughing cycle. Usually a `G70` call to the same `PQ` block definition is used to finish the shape.

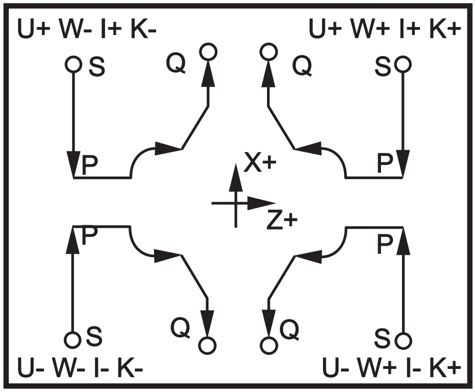
Two types of machining paths are addressed with a `G71` command. The first type of path (Type 1) is when the X-Axis of the programmed path does not change direction. The second type of path (Type 2) allows the X-Axis to change direction. For both Type 1 and Type 2, the programmed path of the Z-axis cannot change direction. If the `P` block contains only an X-Axis position, then Type 1 roughing is assumed. If the `P` block contains both an X-Axis and Z-Axis position, then Type 2 roughing is assumed.

NOTE: *The Z-Axis position given in the `P` block to specify Type 2 roughing does not have to cause axis motion. You can use the current Z-Axis position. For example, In the program example on page **11**, note that the `P1` block (indicated by the comment in parentheses) contains the same Z-Axis position as the start position `G00` block above.*

Any one of the four quadrants of the X-Z plane can be cut by specifying address codes D, I, K, U, and W properly.

In the figures, the start position S is the position of the tool at the time of the G71 call. The Z clearance plane [3] is derived from the Z-axis start position and the sum of W and optional K finish allowance.

F7.19: G71 Address Relationships



Type I Details

When Type I is specified by the programmer it is assumed that the X-axis tool path does not reverse during a cut. Each roughing pass X-axis location is determined by applying the value specified in D to the current X location. The nature of the movement along the Z clearance plane for each roughing pass is determined by the G code in block P. If block P contains a G00 code, then movement along the Z clearance plane is a rapid mode. If block P contains a G01 then movement will be at the G71 feed rate.

Each roughing pass is stopped before it intersects the programmed tool path allowing for both roughing and finishing allowances. The tool is then retracted from the material, at a 45 degree angle. The tool then moves in rapid mode to the Z-axis clearance plane.

When roughing is completed the tool is moved along the tool path to clean up the rough cut. If I and K are specified an additional rough finish cut parallel to the tool path is performed.

Type II Details

When Type II is specified by the programmer the X axis PQ path is allowed to vary (for example, the X-axis tool path can reverse direction).

The X axis PQ path must not exceed the original starting location. The only exception is the ending Q block.

Type II, must have a reference move, in both the X and Z axis, in the block specified by P.

Roughing is similar to Type I except after each pass along the Z axis, the tool will follow the path defined by PQ. The tool will then retract parallel to the X axis. The Type II roughing method does not leave steps in the part prior to finish cutting and typically results in a better finish.

G72 End Face Stock Removal Cycle (Group 00)

First Block (Only use when using two block G72 notation)
*__W__ - Depth of cut for each pass of stock removal, positive radius
*__R__ - Retract height for each pass of stock removal
Second Block
*__D__ - Depth of cut for each pass of stock removal, positive radius (Only use when using one block G72 notation)
*__F__ - Feedrate in inches (mm) per minute (G98) or per revolution (G99) to use throughout G71 PQ block
*__I__ - X-axis size and direction of G72 rough pass allowance, radius
*__K__ - Z-axis size and direction of G72 rough pass allowance
P - Starting Block number of path to rough
Q - Ending Block number of path to rough
*__S__ - Spindle speed to use throughout G72 PQ block
*__T__ - Tool and offset to use throughout G72 PQ block
*__U__ - X-axis size and direction of G72 finish allowance, diameter
*__W__ - Z-axis size and direction of G72 finish allowance

*indicates optional

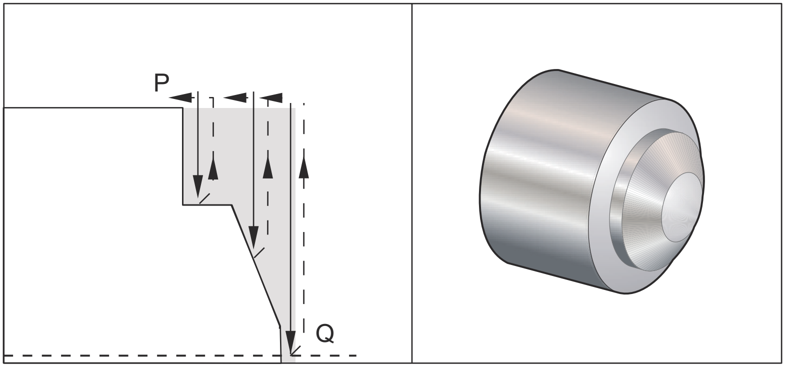
G18 Z-X plane must be active.

2 Block G72 Programming Example:

```
G72 W... R...
G72 F... I... K... P... Q... S... T... U... W...
```

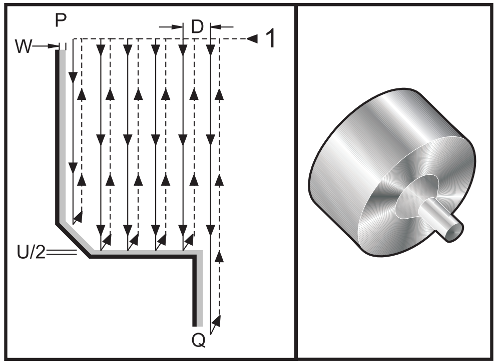
NOTE: *The P values are modal. This means if you are in the middle of a canned cycle and a G04 Pnn or an M97 Pnn is used the P value will be used for the dwell / subprogram as well as the canned cycle.*

F7.20: G72 Basic G Code Example: [P] Starting block, [1] Start position, [Q] Ending block.




```
%
O60721 (G72 END FACE STOCK REMOVAL EX 1) ;
(G54 X0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is an end face cutting tool) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G50 S1000 (Limit spindle to 1000 RPM) ;
G97 S500 M03 (CSS, spindle on CW) ;
G00 G54 X6. Z0.1 (Rapid to clear position) ;
M08 (Coolant on) ;
G96 S200 (CSS on) ;
(BEGIN CUTTING BLOCKS) ;
G72 P1 Q2 D0.075 U0.01 W0.005 F0.012 (Begin G72) ;
N1 G00 Z-0.65 (P1 - Begin toolpath);
G01 X3. F0.006 (1st position);
Z-0.3633 (Face Stock Removal);
X1.7544 Z0. (Face Stock Removal) ;
X-0.0624 ;
N2 G00 Z0.02 (Q2 - End toolpath);
G70 P1 Q2 (Finish Pass) ;
(BEGIN COMPLETION BLOCKS) ;
G97 S500 (CSS off) ;
G00 G53 X0 M09 (X home, coolant off) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;
%
```

F7.21: `G72` Tool Path: [P] Starting block, [1] Start position, [Q] Ending block.



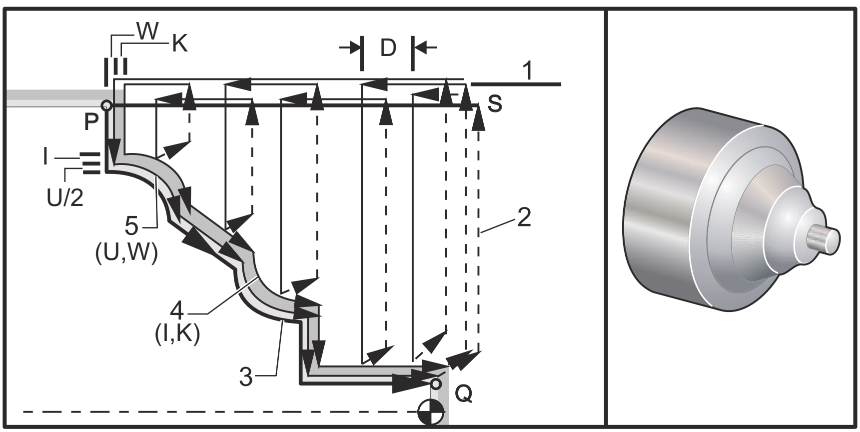

```
%
O60722(G72 END FACE STOCK REMOVAL EX 2) ;
(G54 X0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is an end face cutting tool) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G50 S1000 (Limit spindle to 1000 RPM) ;
G97 S500 M03 (CSS, spindle on CW) ;
G00 G54 X4.05 Z0.2 (Rapid to 1st position) ;
M08 (Coolant on) ;
G96 S200 (CSS on) ;
(BEGIN CUTTING BLOCKS) ;
G72 P1 Q2 U0.03 W0.03 D0.2 F0.01 (Begin G72);
N1 G00 Z-1.(P1 - Begin toolpath) ;
G01 X1.5 (Linear feed) ;
X1. Z-0.75 (Linear feed) ;
G01 Z0 (Linear feed) ;
N2 X0(Q2 - End of toolpath) ;
G70 P1 Q2 (Finishing cycle) ;
(BEGIN COMPLETION BLOCKS) ;
G97 S500 (CSS off) ;
G00 G53 X0 M09 (X home, coolant off) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;
%
```

This canned cycle removes material on a part given the finished part shape. It is similar to G71 but removes material along the face of a part. Define the shape of a part by programming the finished tool path and then use the G72 PQ block. Any F,S or T commands on the G72 line or in effect at the time of the G72 is used throughout the G72 roughing cycle. Usually a G70 call to the same PQ block definition is used to finish the shape.

Two types of machining paths are addressed with a G72 command.

- The first type of path (Type 1) is when the Z Axis of the programmed path does not change direction. The second type of path (Type 2) allows the Z Axis to change direction. For both the first type and the second type of programmed path the X Axis cannot change direction. If Setting 33 is set to FANUC, Type 1 is selected by having only an X-axis motion in the block specified by P in the G72 call.
- When both an X-axis and Z-axis motion are in the P block then Type 2 roughing is assumed.

F7.22: G72 End Face Stock Removal Cycle: [P] Starting block, [1] X-Axis clearance plane, [2] G00 block in P, [3] Programmed path, [4] Roughing allowance, [5] Finishing allowance.

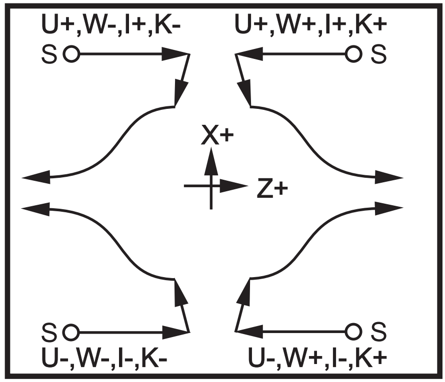


The G72 consists of a roughing phase and a finishing phase. The roughing and finishing phase are handled differently for Type 1 and Type 2. Generally the roughing phase consists of repeated passes along the X-axis at the specified feed rate. The finishing phase consists of a pass along the programmed tool path to remove excess material left by the roughing phase while leaving material for a G70 finishing cycle. The final motion in either type is a return to the starting position S.

In the previous figure the start position S is the position of the tool at the time of the G72 call. The X clearance plane is derived from the X-axis start position and the sum of U and optional I finish allowances.

Any one of the four quadrants of the X-Z plane can be cut by specifying address codes I, K, U, and W properly. The following figure indicates the proper signs for these address codes to obtain the desired performance in the associated quadrants.

F7.23: G72 Address Relationships



G73 Irregular Path Stock Removal Cycle (Group 00)

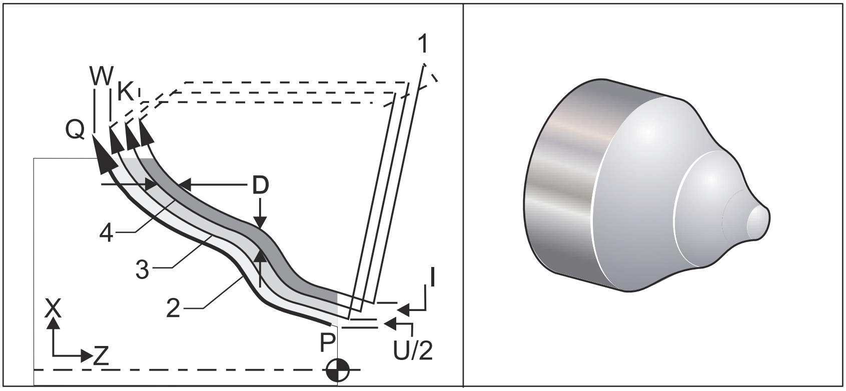
D - Number of cutting passes, positive integer
“**F** - Feedrate in inches (mm) per minute (G98) or per revolution (G99) to use throughout G73 PQ block
I - X-axis distance and direction from first cut to last, radius
K - Z-axis distance and direction from first cut to last
P - Starting Block number of path to rough
Q - Ending Block number of path to rough
*__S__ - Spindle speed to use throughout G73 PQ block
*__T__ - Tool and offset to use throughout G73 PQ block
*__U__ - X-axis size and direction of G73 finish allowance, diameter
*__W__ - Z-axis size and direction of G73 finish allowance

* indicates optional

G18 Z-X plane must be active

NOTE: *The P values are modal. This means if you are in the middle of a canned cycle and a G04 Pnn or an M97 Pnn is used the P value will be used for the dwell / subprogram as well as the canned cycle.*

F7.24: G73 Irregular Path Stock Removal: [P] Starting block, [Q] Ending block [1] Start position, [2] Programmed path, [3] Finish allowance, [4] Roughing allowance.



The G73 canned cycle can be used for rough cutting of preformed material such as castings. The canned cycle assumes that material has been relieved or is missing a certain known distance from the programmed tool path PQ.

Machining starts from the current position (S), and either rapids or feeds to the first rough cut. The nature of the approach move is based on whether a G00 or G01 is programmed in block P. Machining continues parallel to the programmed tool path. When block Q is reached a rapid departure move is executed to the Start position plus the offset for the second roughing pass. Roughing passes continue in this manner for the number of rough passes specified in D. After the last rough is completed, the tool returns to the starting position S.

Only F, S and T prior to or in the G73 block are in effect. Any feed (F), spindle speed (S) or tool change (T) codes on the lines from P to Q are ignored.

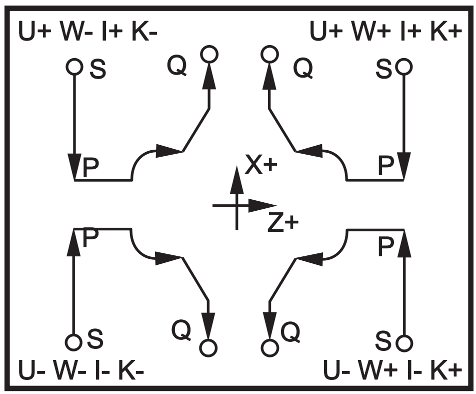
The offset of the first rough cut is determined by (U/2 + I) for the X Axis, and by (W + K) for the Z Axis. Each successive roughing pass moves incrementally closer to the final roughing finish pass by an amount of (I/(D- 1)) in the X Axis, and by an amount of (K/(D-1)) in the Z Axis. The last rough cut always leaves finish material allowance specified by U/2 for the X Axis and W for the Z Axis. This canned cycle is intended for use with the G70 finishing canned cycle.

The programmed tool path PQ does not have to be monotonic in X or Z, but care has to be taken to insure that existing material does not interfere with tool movement during approach and departure moves.

NOTE: *Monotonic curves are curves that tend to move in only one direction as x increases. A monotonic increasing curve always increases as x increases, i.e. f(a)>f(b) for all a>b. A monotonic decreasing curve always decreases as x increases, i.e. f(a)<f(b) for all a>b. The same sort of restrictions are also made for the monotonic non-decreasing and monotonic non-increasing curves.*

The value of `D` must be a positive integral number. If the `D` value includes a decimal, an alarm is generated. The four quadrants of the `ZX` plane can be machined if the following signs for `U`, `I`, `W`, and `K` are used.

F7.25: `G71` Address Relationships

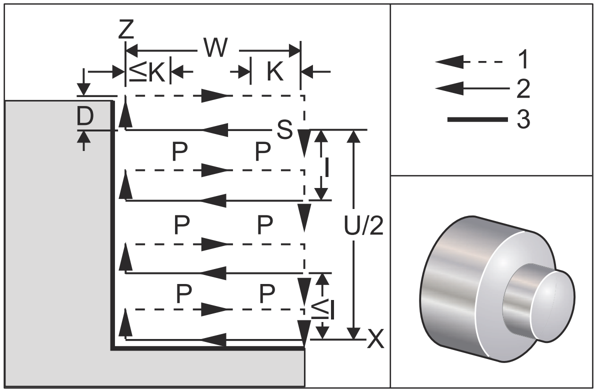


G74 End Face Grooving Cycle (Group 00)

* **D** - Tool clearance when returning to starting plane, positive radius
* **F** - Feed rate
* **I** - X-axis size of increment between peck cycles, positive radius
K - Z-axis size of increment between pecks in a cycle
* **U** - X-axis incremental distance away from current X position before returning to the start plane.
W - Z-axis incremental distance to total pecking depth
X - X-axis absolute location of furthest peck cycle (diameter)
Z - Z-axis absolute location total pecking depth

*indicates optional

F7.26: G74 End Face Grooving Cycle Peck Drilling: [1] Rapid, [2] Feed, [3] Programmed Path, [S] Start position, [P] Peck retraction (Setting 22).



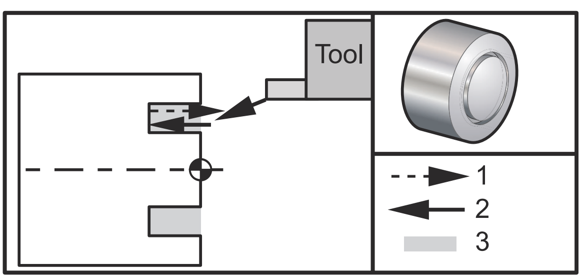
The G74 canned cycle is used for grooving on the face of a part, peck drilling, or turning.

***Warning: The D code command is rarely used and should only be used if the wall on the outside of the groove does not exist like the figure above. The D code can be used in grooving and turning to provide a tool clearance shift, in the X axis, before returning in the Z axis to the “C” clearance point. But, if both sides to the groove exist during the shift, then the groove tool would break. So you wouldn’t want to use the D command.

A minimum of two pecking cycles occur, if an X, or U, code is added to a G74 block and X is not the current position. One at the current location and then at the X location. The I code is the incremental distance between X-Axis pecking cycles. Adding an I performs multiple pecking cycles between the starting position S and X. If the distance between S and X is not evenly divisible by I then the last interval is less than I.

When K is added to a G74 block, pecking is performed at each interval specified by K, the peck is a rapid move opposite the direction of feed with a distance defined by Setting 22. The D code can be used for grooving and turning to provide material clearance when returning to starting plane S.

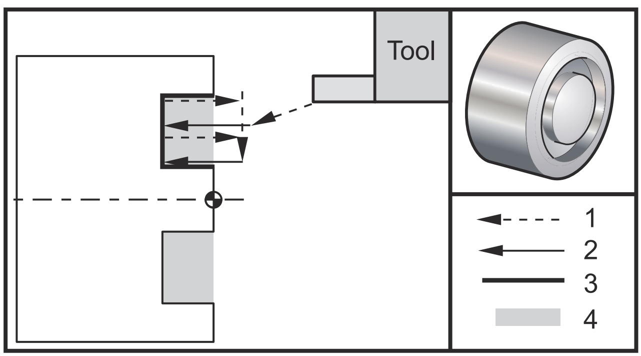
F7.27: G74 End Face Grooving Cycle: [1] Rapid, [2] Feed, [3] Groove.




```
%
O60741 (G74 END FACE) ;
(G54 X0 is at the center of rotation) ;
```

```
(Z0 is on the face of the part) ;
(T1 is an end face cutting tool) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G50 S1000 (Limit spindle to 1000 RPM) ;
G97 S500 M03 (CSS off, Spindle on CW) ;
G00 G54 X3. Z0.1 (Rapid to 1st position) ;
M08 (Coolant on) ;
G96 S200 (CSS on) ;
(BEGIN CUTTING BLOCKS) ;
G74 Z-0.5 K0.1 F0.01 (Begin G74) ;
(BEGIN COMPLETION BLOCKS) ;
G97 S500 (CSS off) ;
G00 G53 X0 M09 (X home, coolant off) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;
%
```

F7.28: `G74` End Face Grooving Cycle (Multiple Pass): [1] Rapid, [2] Feed, [3] Programmed path, [4] Groove.




```
%
O60742 (G74 END FACE MULTI PASS) ;
(G54 X0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is an end face cutting tool) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G50 S1000 (Limit spindle to 1000 RPM) ;
G97 S500 M03 (CSS off, spindle on CW) ;
G00 G54 X3. Z0.1 (Rapid to 1st position) ;
```

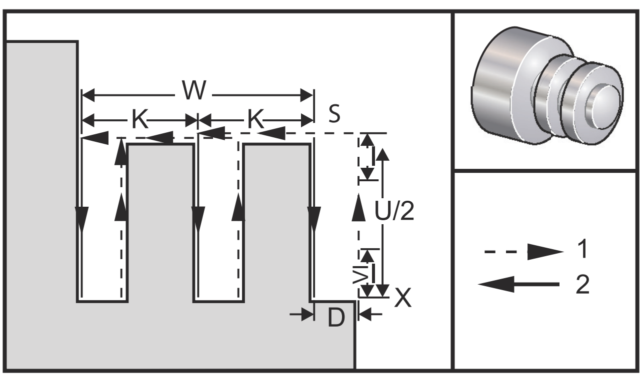
```
M08 (Coolant on) ;
G96 S200 (CSS on) ;
(BEGIN CUTTING BLOCKS) ;
G74 X1.75 Z-0.5 I0.2 K0.1 F0.01 (Begin G74) ;
(BEGIN COMPLETION BLOCKS) ;
G97 S500 (CSS off) ;
G00 G53 X0 M09 (X home, coolant off) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;
%
```

G75 O.D./I.D. Grooving Cycle (Group 00)

*__D__ - Tool clearance when returning to starting plane, positive
*__F__ - Feed rate
*__I__ - X-axis size of increment between pecks in a cycle (radius measure)
*__K__ - Z-axis size of increment between peck cycles
*__U__ - X-axis incremental distance to total pecking depth
W - Z-axis incremental distance to furthest peck cycle
X - X-axis absolute location total pecking depth (diameter)
Z - Z-axis absolute location to furthest peck cycle

* indicates optional

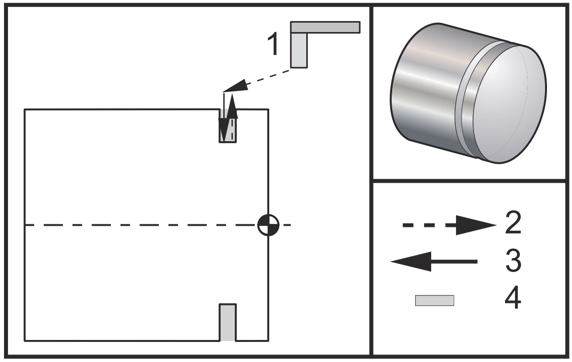
F7.29: G75 O.D./I.D. Grooving Cycle: [1] Rapid, [2] Feed, [S] Start position.



The G75 canned cycle can be used for grooving an outside diameter. When a Z, or W, code is added to a G75 block and Z is not the current position, then a minimum of two pecking cycles occur. One at the current location and another at the Z location. The K code is the incremental distance between Z axis pecking cycles. Adding a K performs multiple, evenly spaced, grooves. If the distance between the starting position and the total depth (Z) is not evenly divisible by K then the last interval along Z is less than K.

NOTE: *Chip clearance is defined by Setting 22.*

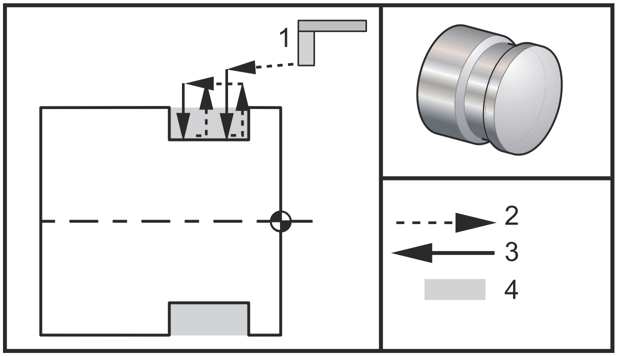
F7.30: `G75` O.D. Single Pass




```
%
O60751 (G75 OD GROOVE CYCLE) ;
(G54 X0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is an OD groove tool) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G50 S1000 (Limit spindle to 1000 RPM) ;
G97 S500 M03 (CSS off, spindle on CW) ;
G00 G54 X4.1 Z0.1 (Rapid to 1st position) ;
M08 (Coolant on) ;
G96 S200 (CSS on) ;
(BEGIN CUTTING BLOCKS) ;
G01 Z-0.75 F0.05 (Feed to Groove location) ;
G75 X3.25 I0.1 F0.01 (Begin G75) ;
(BEGIN COMPLETION BLOCKS) ;
G97 S500 (CSS off) ;
G00 G53 X0 M09 (X home, coolant off) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;
%
```

The following program is an example of a `G75` program (Multiple Pass):

F7.31: G75 O.D. Multiple Pass: [1] Tool, [2] Rapid, [3] Feed, [4] Groove.




```
%
O60752 (G75 OD GROOVE CYCLE 2) ;
(G54 X0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is an OD groove tool) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G50 S1000 (Limit spindle to 1000 RPM) ;
G97 S500 M03 (CSS off, spindle on CW) ;
G00 G54 X4.1 Z0.1 (Rapid to 1st position) ;
M08 (Coolant on) ;
G96 S200 (CSS on) ;
(BEGIN CUTTING BLOCKS) ;
G01 Z-0.75 F0.05 (Feed to Groove location) ;
G75 X3.25 Z-1.75 I0.1 K0.2 F0.01 (Begin G75) ;
(BEGIN COMPLETION BLOCKS) ;
G97 S500 (CSS off) ;
G00 G53 X0 M09 (X home, coolant off) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;
%
```